IBM Developer
SKILLS NETWORK

IBM Data Science Professional Certificate

Winning Space Race with Data Science

Your Name
October 15, 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Q3

Summary of methodologies

- Collected 90 Falcon 9 launches from the SpaceX API and Wikipedia scraping
- Wrangled the data and created a landing-success label
- Explored it with SQL and visual charts
- Built a Folium map and a Plotly Dash dashboard
- Tuned four classifiers (LR, SVM, Decision Tree, KNN) with GridSearchCV

Summary of all results

66.7%
landing rate

76.9%
best site: KSC

87%
best model CV

- Success climbed from 0% (2010–2013) to about 85–90% by 2019–2020
- Orbit matters: GTO is hardest at about 52%
- All four models reach 83% test accuracy; the Decision Tree wins on cross-validation

Introduction

Q4

Project background and context

- SpaceX advertises Falcon 9 launches at about $62 million, versus $165 million or more from other providers
- Most of the saving comes from reusing the first stage, so a landing that succeeds lowers the price of the next launch
- If we can predict whether the first stage will land, we can estimate the real cost of a launch
- This matters to a competing company (SpaceY) that wants to bid against SpaceX

Problems we want to answer

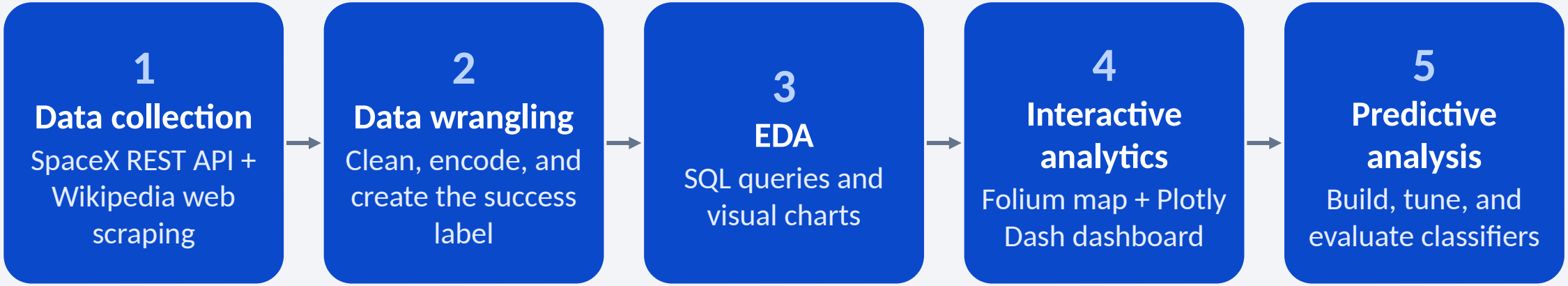
- Which factors (payload, orbit, launch site, booster version, flight number) drive landing success?
- Does the success rate improve with experience over the years?
- Which launch sites are the most successful, and where are they placed relative to coastlines, railways and highways?
- Can a classification model predict first-stage landing success accurately enough to estimate launch cost?

Section 1

Methodology



Methodology



- **Data collection:** REST calls to api.spacexdata.com and BeautifulSoup scraping of the Falcon 9 launch history table
- **Wrangling:** missing payload mass replaced by the mean; outcomes mapped to Class = 1 (landed) or 0 (did not land)
- **Modelling:** standardize features, 80/20 split, GridSearchCV with 10-fold cross-validation, compare accuracy and confusion matrices

Data Collection

Q5

Source 1: SpaceX REST API

- api.spacexdata.com/v4
- Extra calls resolve rocket, pad, payload and core IDs
- Gives booster, site, payload mass, orbit, outcome

Source 2: Wikipedia scraping

- Falcon 9 launch history page (static snapshot)
- BeautifulSoup extracts the tables into a DataFrame
- Adds flight number, date, customer, outcome

Combined dataset

- 90 Falcon 9 launches
- 17 features per launch
- Saved as CSV files for the next steps

API request
requests.get()

→

Parse JSON
json_normalize

→

Scrape HTML
BeautifulSoup

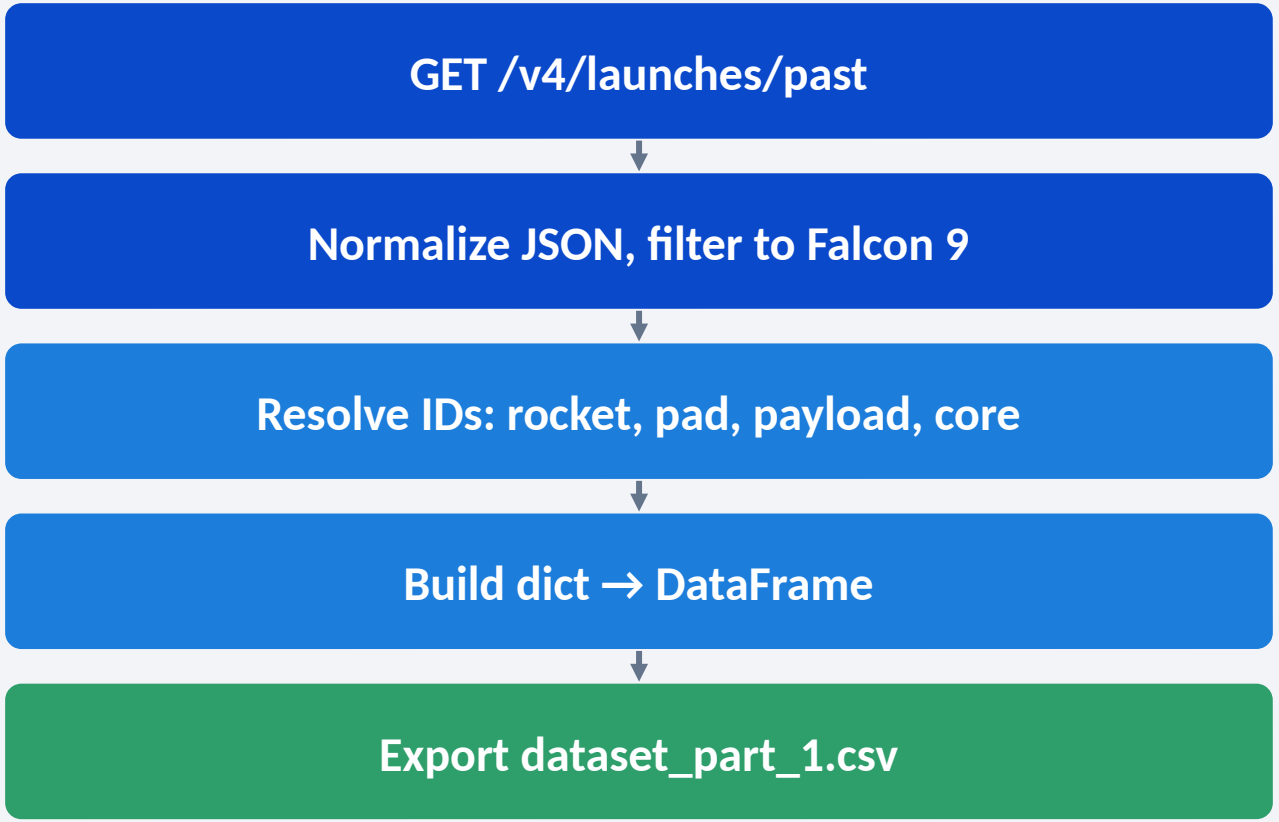
→

Merge + save
CSV files

Data Collection – SpaceX API

Q5

- Request past launches with requests.get() on the SpaceX v4 API and decode the JSON with .json()
- Keep only single-core, single-payload Falcon 9 launches up to 13 Nov 2020
- Make extra API calls to turn the rocket, launchpad, payload and core IDs into names, orbit, mass and landing results
- Build a dictionary, convert it to a pandas DataFrame, and export to CSV (90 rows)

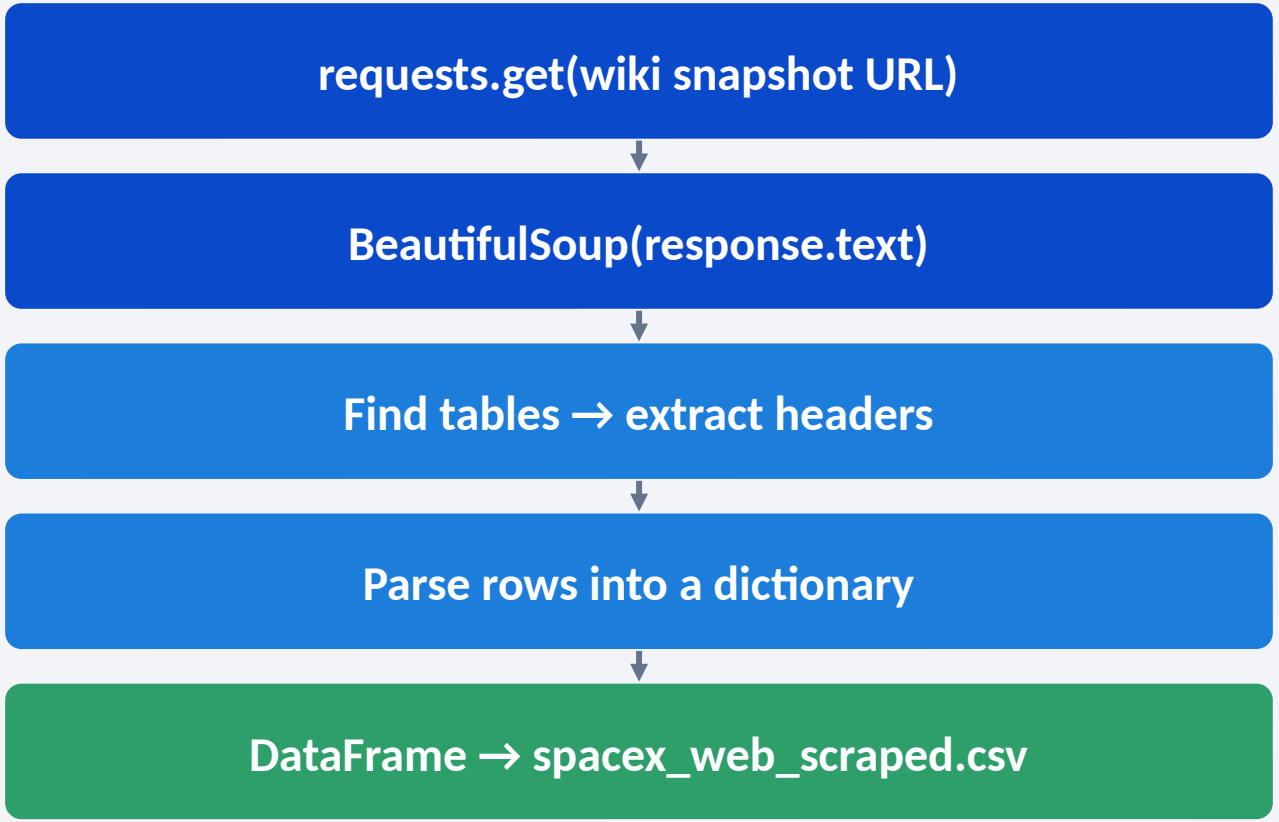


GitHub: https://github.com/your-username/ds-capstone-spacex/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

Data Collection - Scraping

Q5

- Send an HTTP GET request to a static snapshot of the Wikipedia “List of Falcon 9 and Falcon Heavy launches” page
- Parse the HTML with BeautifulSoup and select all tables
- Extract column names from the launch table header
- Loop through each row, clean the cells, and fill a dictionary with flight number, date, booster, site, payload, orbit, customer and outcome
- Convert to a DataFrame and export to CSV



GitHub: https://github.com/your-username/ds-capstone-spacex/blob/main/jupyter-labs-webscraping.ipynb

Data Wrangling

Q5

90	3	11	66.7%
launches	launch sites	orbit types	success rate



- **Bad outcomes** (Class = 0): False ASDS, False Ocean, False RTLS, None ASDS, None None. **Good outcomes** (Class = 1): True ASDS, True Ocean, True RTLS.
- Missing payload mass replaced with the column mean; the mean of Class gives the 66.7% success rate

GitHub: https://github.com/your-username/ds-capstone-spacex/blob/main/labs-jupyter-spacex-Data-wrangling.ipynb

EDA with Data Visualization

Q6

Chart	Variables plotted	Why this chart
Scatter	Flight Number vs. Launch Site	Shows how success changes with experience at each site
Scatter	Payload vs. Launch Site	Checks whether heavy payloads fly from specific sites
Bar	Success rate vs. Orbit type	Compares success rate across categories at a glance
Scatter	Flight Number vs. Orbit, Payload vs. Orbit	Reveals orbit-specific patterns in the outcome
Line	Average success rate by year	Shows the trend over time
Feature engineering	One-hot encoding of Orbit, LaunchSite, LandingPad, Serial	Prepares numeric features for the models

GitHub: https://github.com/your-username/ds-capstone-spacex/blob/main/edadataviz.ipynb

EDA with SQL

Q6

- Displayed the unique launch site names
- Listed 5 records where the site name begins with ‘CCA’
- Summed the payload mass carried for NASA (CRS)
- Averaged the payload mass of booster version F9 v1.1
- Found the first successful ground-pad landing date
- Listed drone-ship successes with payload between 4,000 and 6,000 kg
- Counted successful and failed mission outcomes
- Found the boosters that carried the maximum payload (subquery)
- Listed 2015 drone-ship landing failures with booster and site
- Ranked landing outcomes between 2010-06-04 and 2017-03-20

GitHub: https://github.com/your-username/ds-capstone-spacex/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

Q6

Markers + labels

folium.Marker with a DivIcon text label at each of the 4 launch sites

Why: Pinpoint every site on the map

Circles

folium.Circle (blue) with a radius around each launch site

Why: Highlight the area around each site

Marker cluster

MarkerCluster of green (success) and red (failure) markers for every launch

Why: Show outcomes per site without clutter

Lines + distance labels

MousePosition for coordinates, PolyLine to the nearest coast, railway, highway, city

Why: Measure proximity to infrastructure

GitHub: https://github.com/your-username/ds-capstone-spacex/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

Q6

Inputs
Launch-site dropdown, payload range slider

→

Callbacks
@app.callback filters the dataframe

→

Outputs
Success pie chart, payload scatter chart

Dropdown

Select ALL sites or one site

Why: compare sites quickly

Pie chart

Success count per site, or success vs. failure for a chosen site

Why: shows each site's share

Slider + scatter

Payload range slider drives a Payload vs. Outcome scatter coloured by booster version

Why: link payload and booster to success

GitHub: https://github.com/your-username/ds-capstone-spacex/blob/main/spacex_dash_app.py

Predictive Analysis (Classification)

Q7

Load data
dataset_part_2/3

→

Standardize
StandardScaler

→

Split 80/20
72 train, 18 test

→

GridSearchCV
cv = 10, 4 models

→

Evaluate
accuracy + confusion matrix

→

Select best
highest accuracy

Model	Hyperparameters tuned
Logistic Regression	C = 0.01, 0.1, 1; penalty = l2; solver = lbfgs
Support Vector Machine	kernel (linear, rbf, poly, sigmoid), C = 10^-3 to 10^3, gamma = 10^-3 to 10^3
Decision Tree	criterion, splitter, max_depth 2–18, max_features, min_samples_leaf / split
K-Nearest Neighbours	n_neighbors 1–10, algorithm, p = 1 or 2

GitHub: https://github.com/your-username/ds-capstone-spacex/blob/main/SpaceX_Machine_Learning_Prediction_Part_5.ipynb

Results

Exploratory data analysis

- Success rises with flight number and year
- Orbit type changes success odds a lot
- KSC LC-39A and VAFB SLC-4E lead on success rate

Interactive analytics

- Folium: sites sit near the coast, away from cities
- Dash: KSC LC-39A has the best success ratio
- Payloads of about 2,000–5,500 kg land most often

Predictive analysis

- Four models tested; all reach 83% test accuracy
- Decision Tree has the best cross-validation score (87%)
- Zero missed successful landings; 3 false alarms

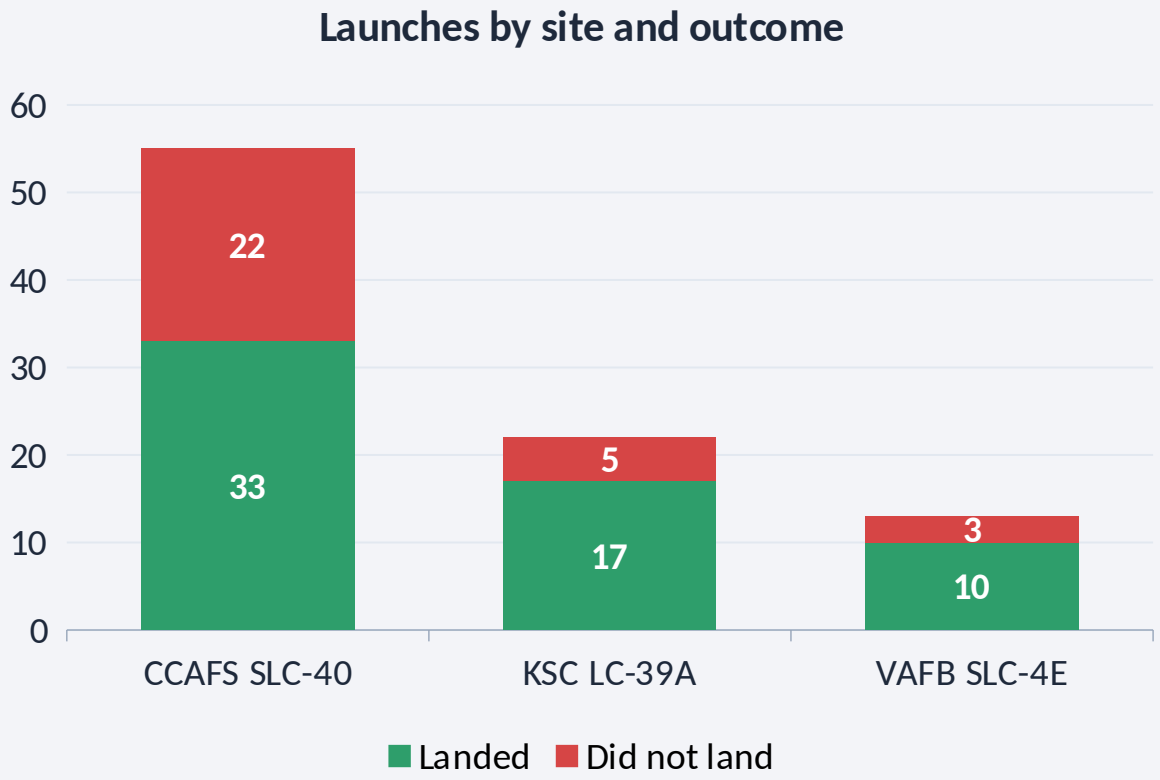
Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

Q8

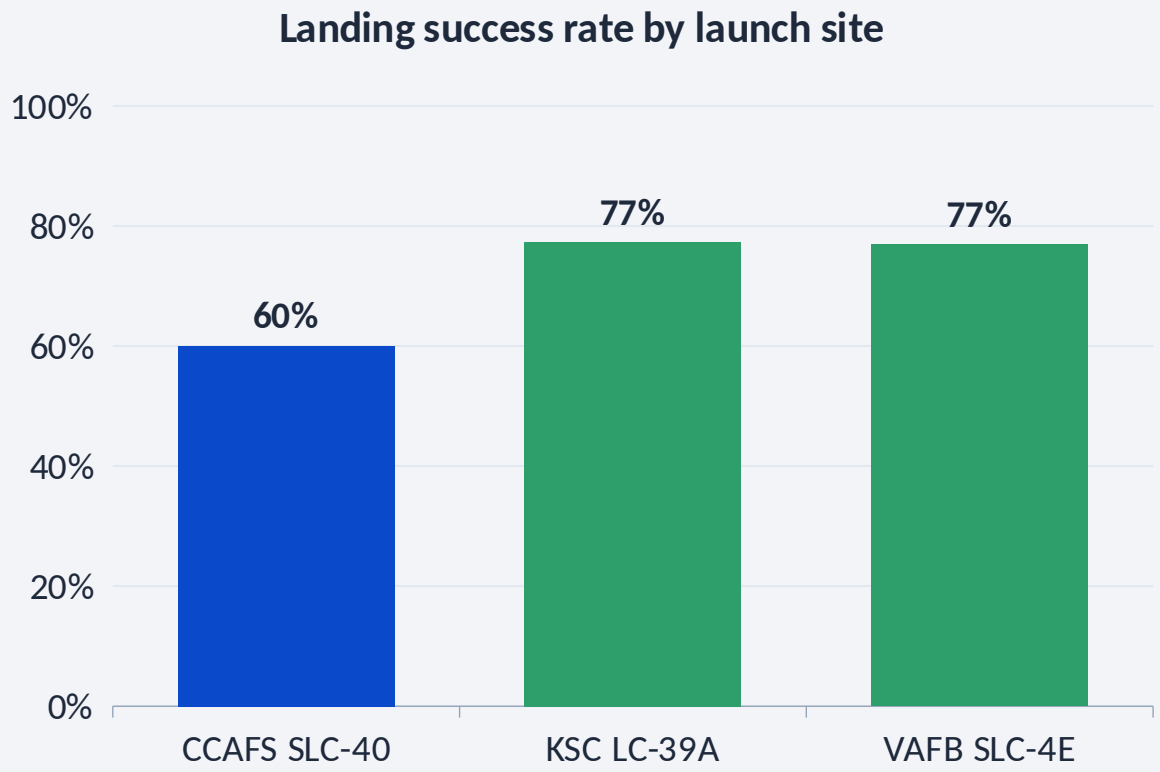
- **Plot:** scatter of Flight Number (x) vs. Launch Site (y), coloured by Class
- Early flights (low flight numbers) at CCAFS SLC-40 are mostly failures
- Later flights are mostly successes at every site
- CCAFS SLC-40 hosts 55 of the 90 launches, the most of any site
- **Insight:** more flights at a site goes with a higher success rate



Payload vs. Launch Site

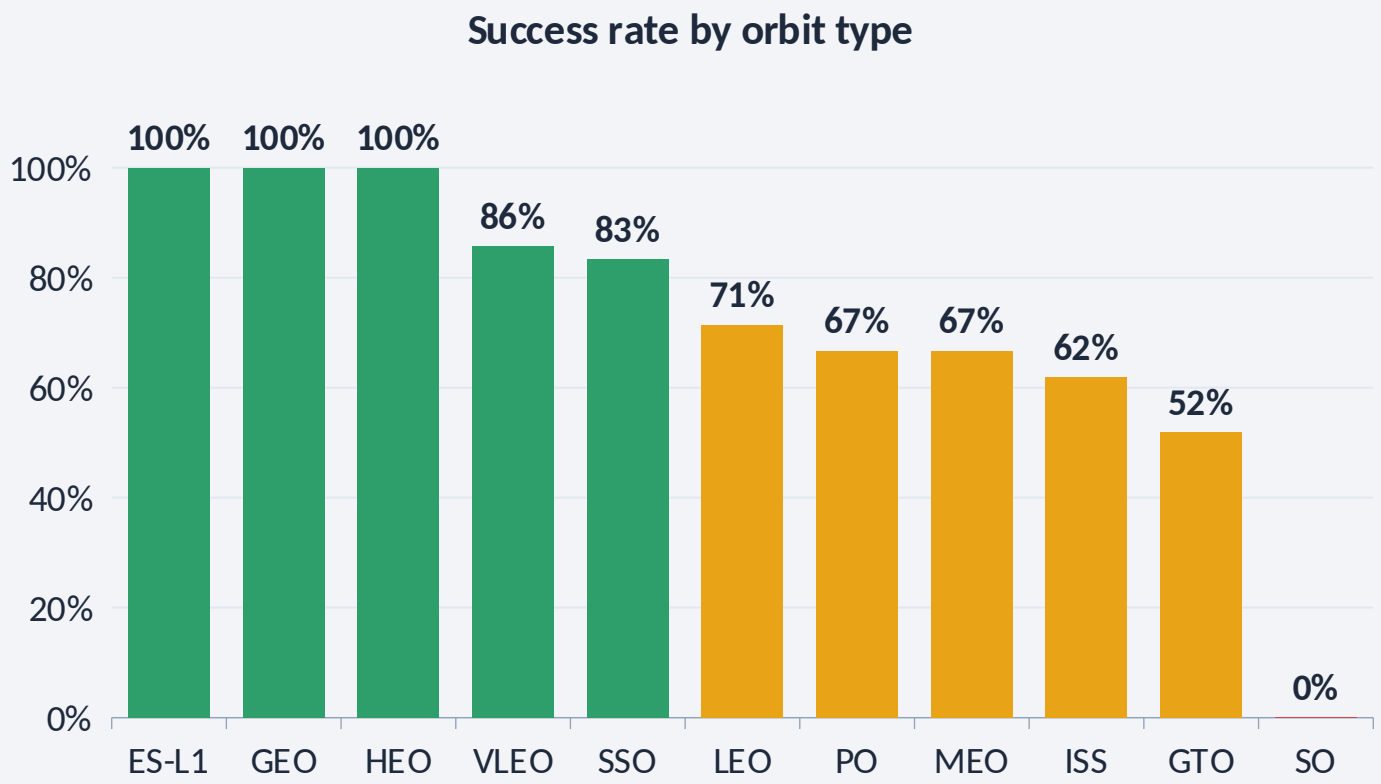
Q8

- **Plot:** scatter of Payload Mass (x) vs. Launch Site (y), coloured by Class
- VAFB SLC-4E launched no payload heavier than about 10,000 kg
- Heavy payloads (above 10,000 kg) fly mainly from CCAFS SLC-40 and KSC LC-39A
- KSC LC-39A and VAFB SLC-4E land about 77% of the time; CCAFS SLC-40 lands 60%
- **Insight:** payload alone does not decide the site; the site mix is what shapes the success rate



Success Rate vs. Orbit Type

Q8

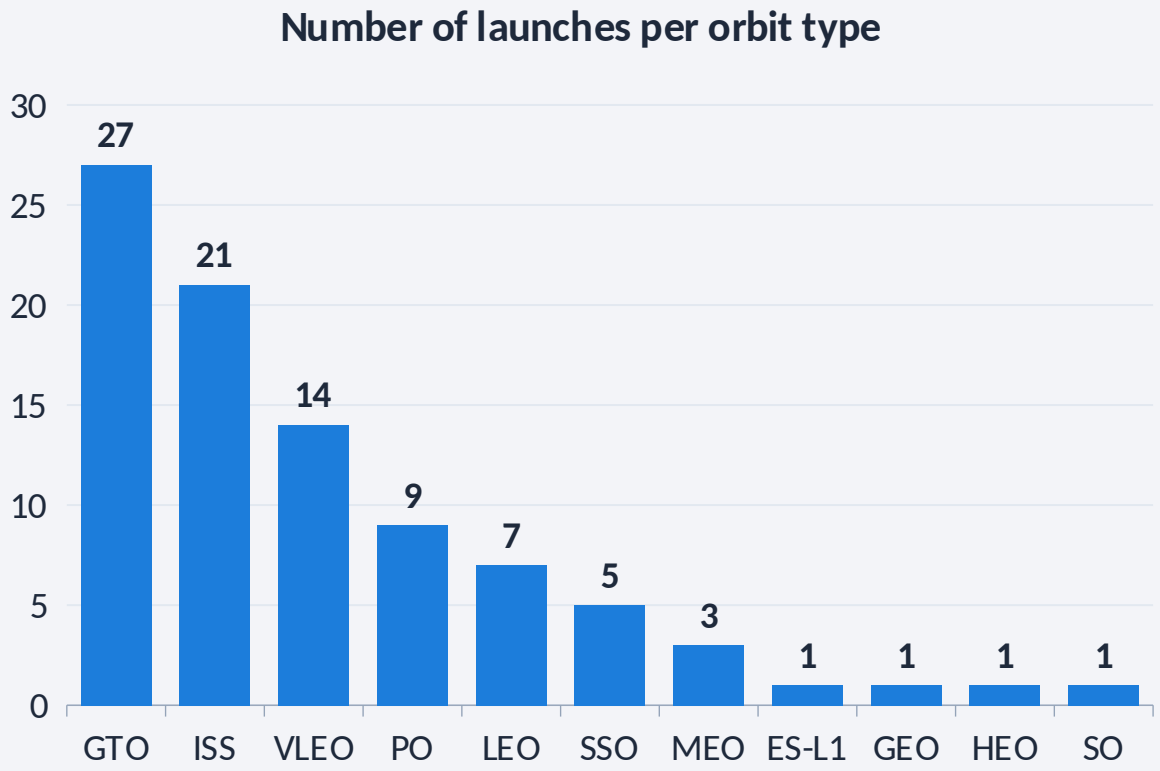


- ES-L1, GEO and HEO landed on every flight, but each has very few launches
- SSO and VLEO land more than 80% of the time
- GTO is the hardest orbit at about 52%, since high-energy missions leave less fuel for landing
- SO never landed in the data

Flight Number vs. Orbit Type

Q8

- **Plot:** scatter of Flight Number (x) vs. Orbit (y), coloured by Class
- In LEO, success rises with flight number, so experience helps
- In GTO there is no clear link between flight number and success
- VLEO flights appear only at high flight numbers, with mostly successful landings
- **Insight:** orbit mix changed over time; later missions favour easier orbits



Payload vs. Orbit Type

Q8

0 – 15,600 kg
payload range in the data

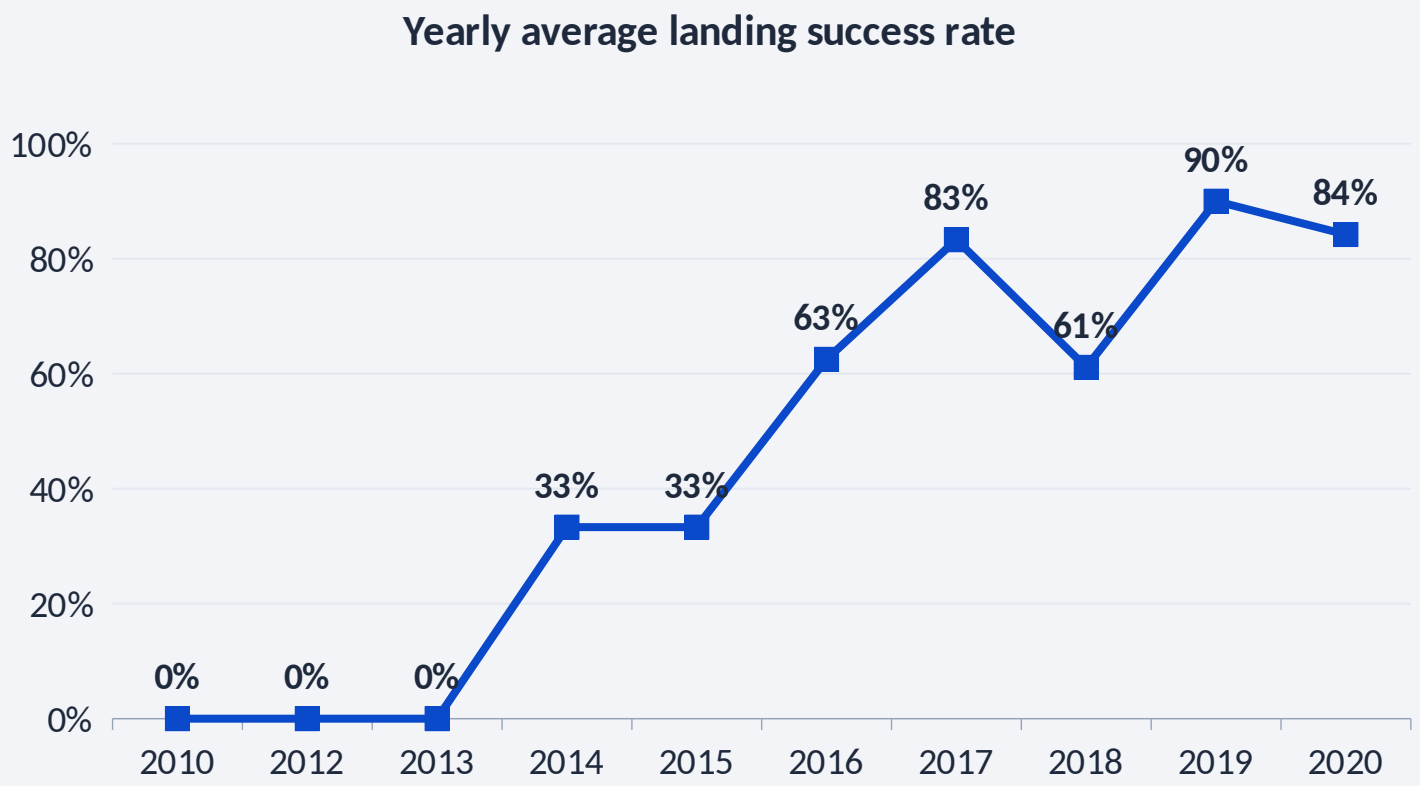
PO · LEO · ISS
heavy payloads land well

GTO
mixed outcomes at any payload

- **Plot:** scatter of Payload Mass (x) vs. Orbit (y), coloured by Class
- With heavy payloads, successful landings are more frequent in Polar, LEO and ISS orbits
- For GTO, successful and failed landings overlap, so payload mass does not separate the classes there
- Light payloads dominate ISS resupply missions, while heavy Starlink-style loads go to LEO
- **Insight:** a model needs orbit and payload together; neither feature is enough alone

Launch Success Yearly Trend

Q8



- No successful landings from 2010 to 2013
- Success starts in 2014 and climbs quickly to about 83% in 2017
- A dip in 2018 is followed by 90% in 2019
- **Insight:** about 85–90% of recent launches land, so reuse is now routine

All Launch Site Names

Q9

```
SELECT DISTINCT Launch_Site
FROM SPACEXTBL;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Explanation

DISTINCT removes duplicates, leaving 4 unique sites: three in Florida (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A) and one in California (VAFB SLC-4E). CCAFS LC-40 was later renamed SLC-40, which is why both names appear.

Launch Site Names Begin with 'CCA'

Q9

```
SELECT * FROM SPACEXTBL
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5;
```

Date	Booster	Payload	Mass (kg)	Customer	Landing outcome
2010-06-04	F9 v1.0 B0003	Dragon Qualification Unit	0	SpaceX	Failure (parachute)
2010-12-08	F9 v1.0 B0004	Dragon demo flight C1	0	NASA (COTS)	Failure (parachute)
2012-05-22	F9 v1.0 B0005	Dragon demo flight C2	525	NASA (COTS)	No attempt
2012-10-08	F9 v1.0 B0006	SpaceX CRS-1	500	NASA (CRS)	No attempt
2013-03-01	F9 v1.0 B0007	SpaceX CRS-2	677	NASA (CRS)	No attempt

Explanation

LIKE 'CCA%' matches every site starting with CCA. The first Falcon 9 launches carried Dragon test and cargo missions and had no landing attempts yet.

Total Payload Mass

Q9

```
SELECT SUM(PAYLOAD_MASS__KG_) AS total_payload_kg
FROM SPACEXTBL
WHERE Customer = 'NASA (CRS)';
```

total_payload_kg
45,596

45,596 kg
carried for NASA (CRS)

Explanation

SUM adds the payload mass of all boosters launched for NASA's Commercial Resupply Services. That is about 45.6 tonnes delivered to the ISS.

Average Payload Mass by F9 v1.1

Q9

```
SELECT AVG(PAYLOAD_MASS__KG_) AS avg_payload_kg
FROM SPACEXTBL
WHERE Booster_Version = 'F9 v1.1';
```

avg_payload_kg
2,928.4

2,928 kg

average for F9 v1.1

Explanation

AVG on the F9 v1.1 booster version gives about 2,928 kg per launch. This early booster flew relatively light payloads compared with later Block 5 boosters.

First Successful Ground Landing Date

Q9

```
SELECT MIN(Date) AS first_ground_landing
FROM SPACEXTBL
WHERE Landing_Outcome = 'Success (ground pad)';
```

first_ground_landing
2015-12-22

2015-12-22

first ground-pad landing

Explanation

MIN(Date) returns the earliest success on a ground pad: 22 December 2015, the first time a Falcon 9 first stage landed back on land.

Successful Drone Ship Landing with Payload between 4000 and 60

Q9

```
SELECT DISTINCT Booster_Version
FROM SPACEXTBL
WHERE Landing_Outcome = 'Success (drone ship)'
  AND PAYLOAD_MASS__KG_ > 4000
  AND PAYLOAD_MASS__KG_ < 6000;
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Explanation

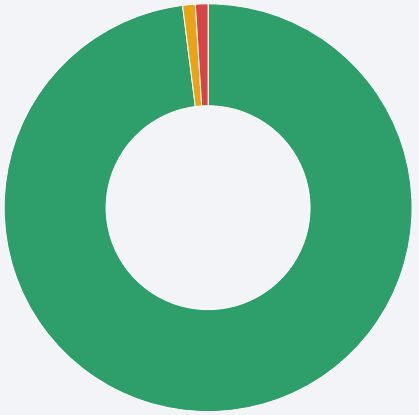
Four Full Thrust boosters landed on a drone ship while carrying between 4,000 and 6,000 kg. All of them are F9 FT, which shows the FT version handled mid-weight payloads well.

Total Number of Successful and Failure Mission Outcomes

Q9

```
SELECT Mission_Outcome, COUNT(*) AS total
FROM SPACEXTBL
GROUP BY Mission_Outcome;
```

Mission_Outcome	total
Success	98
Success (payload status unclear)	1
Failure (in flight)	1



Explanation

99 of 100 missions succeeded (98 clear + 1 unclear). Only one mission failed in flight.

Boosters Carried Maximum Payload

Q9

```
SELECT DISTINCT Booster_Version
FROM SPACEXTBL
WHERE PAYLOAD_MASS__KG_ =
  (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL);
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4

Booster_Version
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

15,600 kg
maximum payload

Explanation

A subquery finds the maximum payload (15,600 kg). Twelve boosters carried it, and all are Block 5 (F9 B5), the latest, most reusable version.

2015 Launch Records

Q9

```
SELECT substr(Date,6,2) AS month, Date, Booster_Version,
       Launch_Site, Landing_Outcome
FROM SPACEXTBL
WHERE Landing_Outcome = 'Failure (drone ship)'
  AND substr(Date,0,5) = '2015';
```

month	Date	Booster_Version	Launch_Site	Landing_Outcome
10	2015-10-01	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	2015-04-14	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Explanation

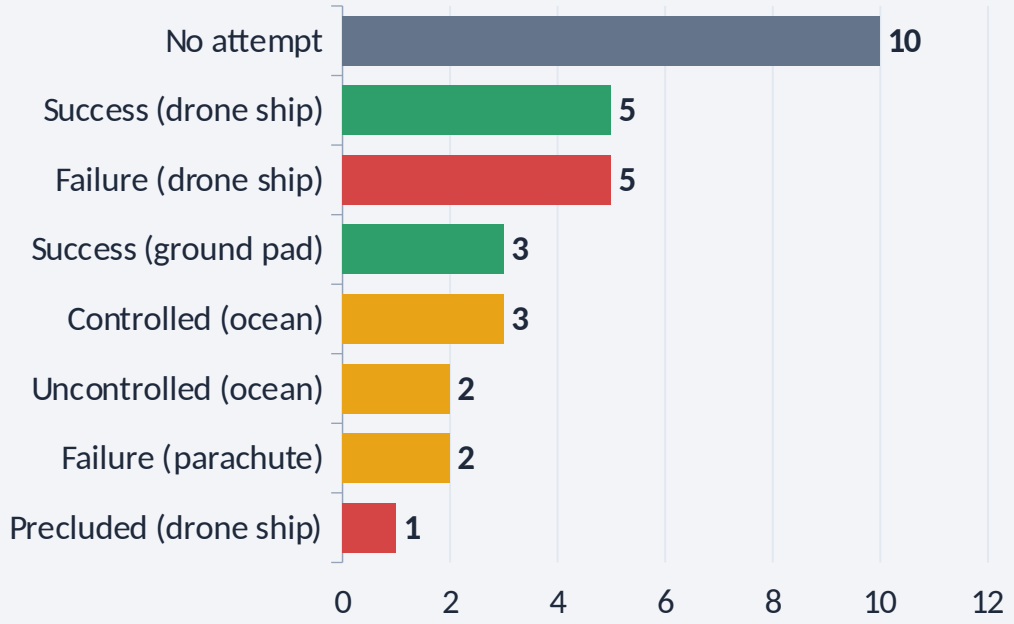
Two early drone-ship landing attempts failed in 2015. Both used the F9 v1.1 booster from CCAFS LC-40, consistent with the low early success rates seen in the yearly trend.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-2

Q9

```
SELECT Landing_Outcome, COUNT(*) AS cnt
FROM SPACEXTBL
WHERE Date BETWEEN '2010-06-04'
  AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY cnt DESC;
```

Landing_Outcome	cnt
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1



Explanation

10 launches made no landing attempt. Drone-ship successes and failures tie at 5 each, but ground-pad landings succeeded 3 of 3.

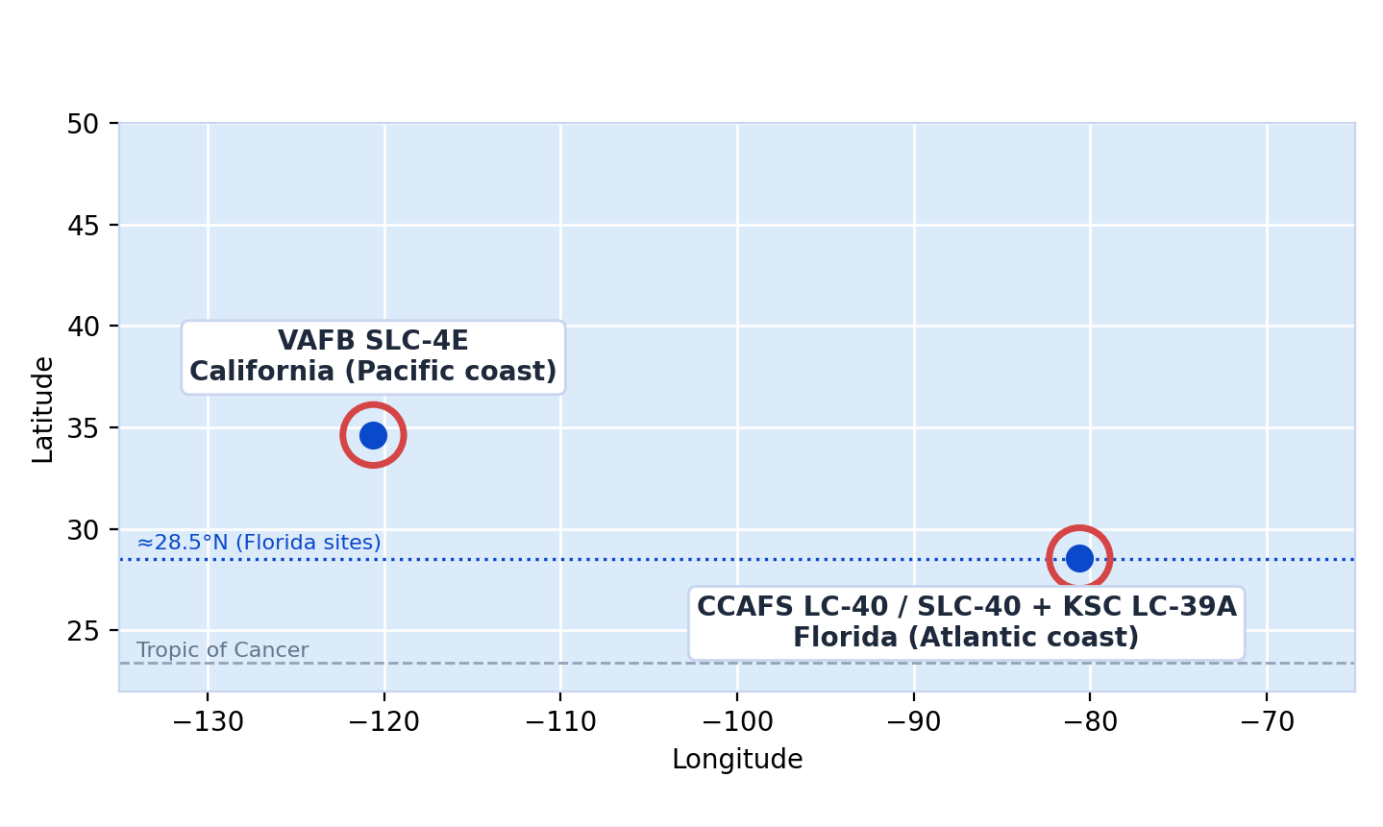


Section 3

Launch Sites Proximities Analysis

Launch Site Locations on the Global Map

Q10

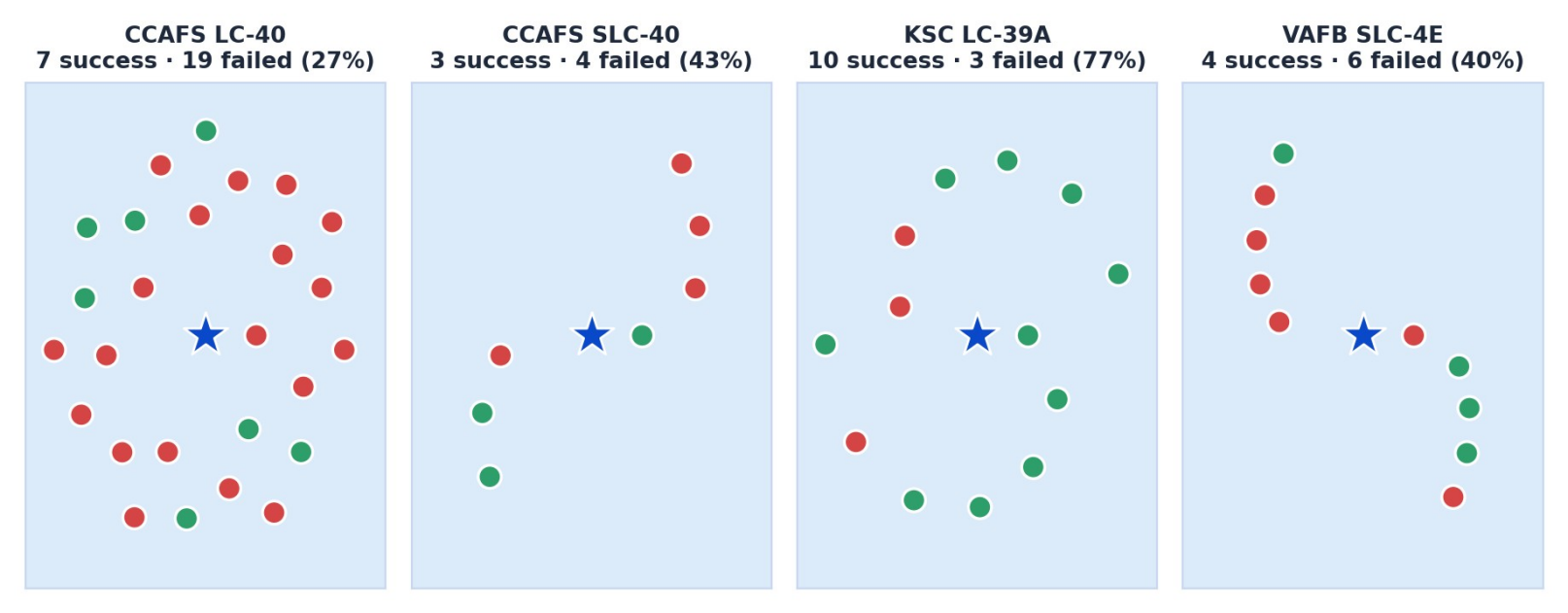


Schematic plot of the exact site coordinates used in the Folium map.

- Markers show all four launch sites (3 in Florida, 1 in California)
- All sites sit close to the coast, so failed or aborted stages fall over water
- Florida sites are near the equator (≈28.5°N), where Earth's rotation adds speed for launches
- VAFB is on the Pacific coast and suits polar and sun-synchronous orbits

Colour-Labelled Launch Outcomes by Site

Q10



Green = successful landing

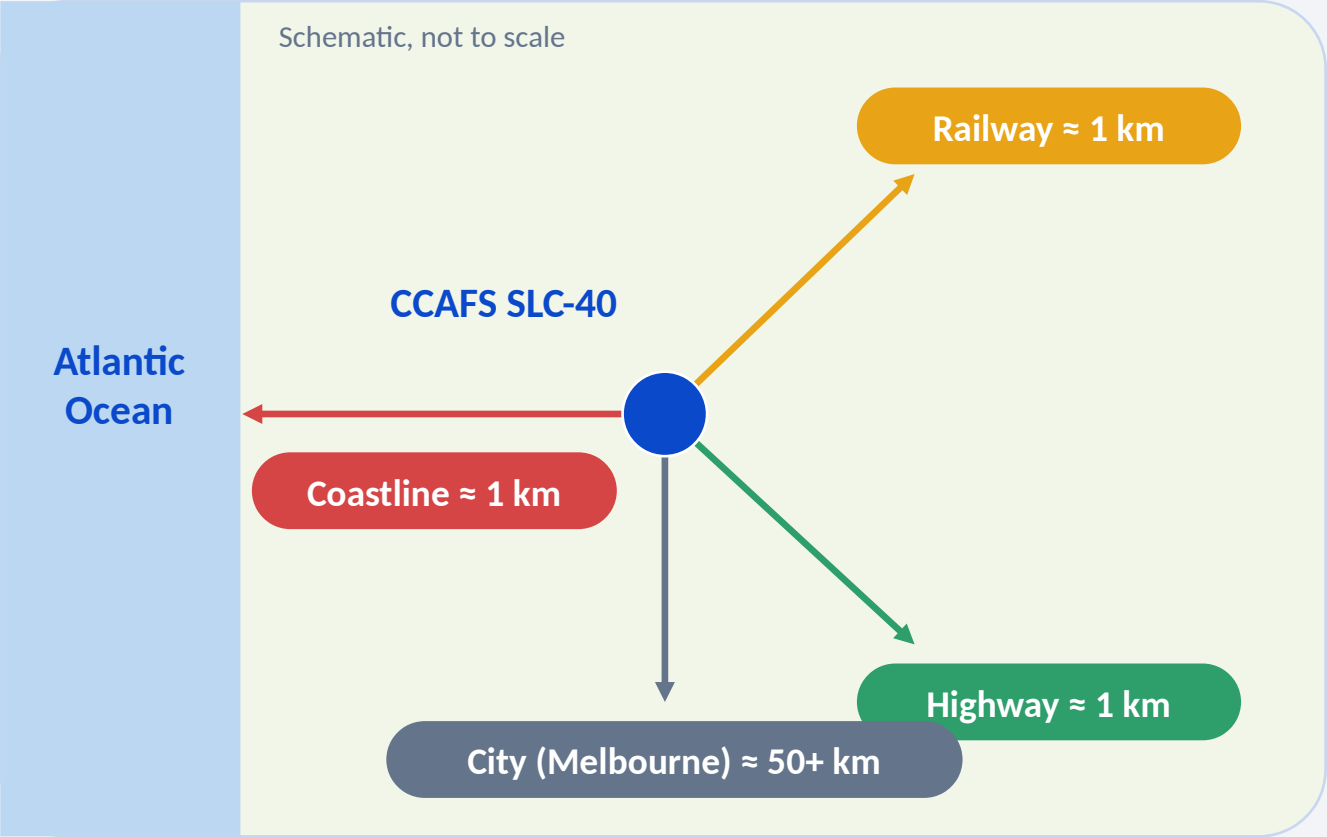
Red = failed landing

- KSC LC-39A has the most green markers: 10 of 13
- CCAFS LC-40 has many red markers: 19 of 26 failed

- Marker clusters group the 56 launch records at each site; colour makes the best and worst sites obvious at a glance. KSC LC-39A is the standout at 77% success.

Launch Site Proximity to Coast, Rail, Highway and City

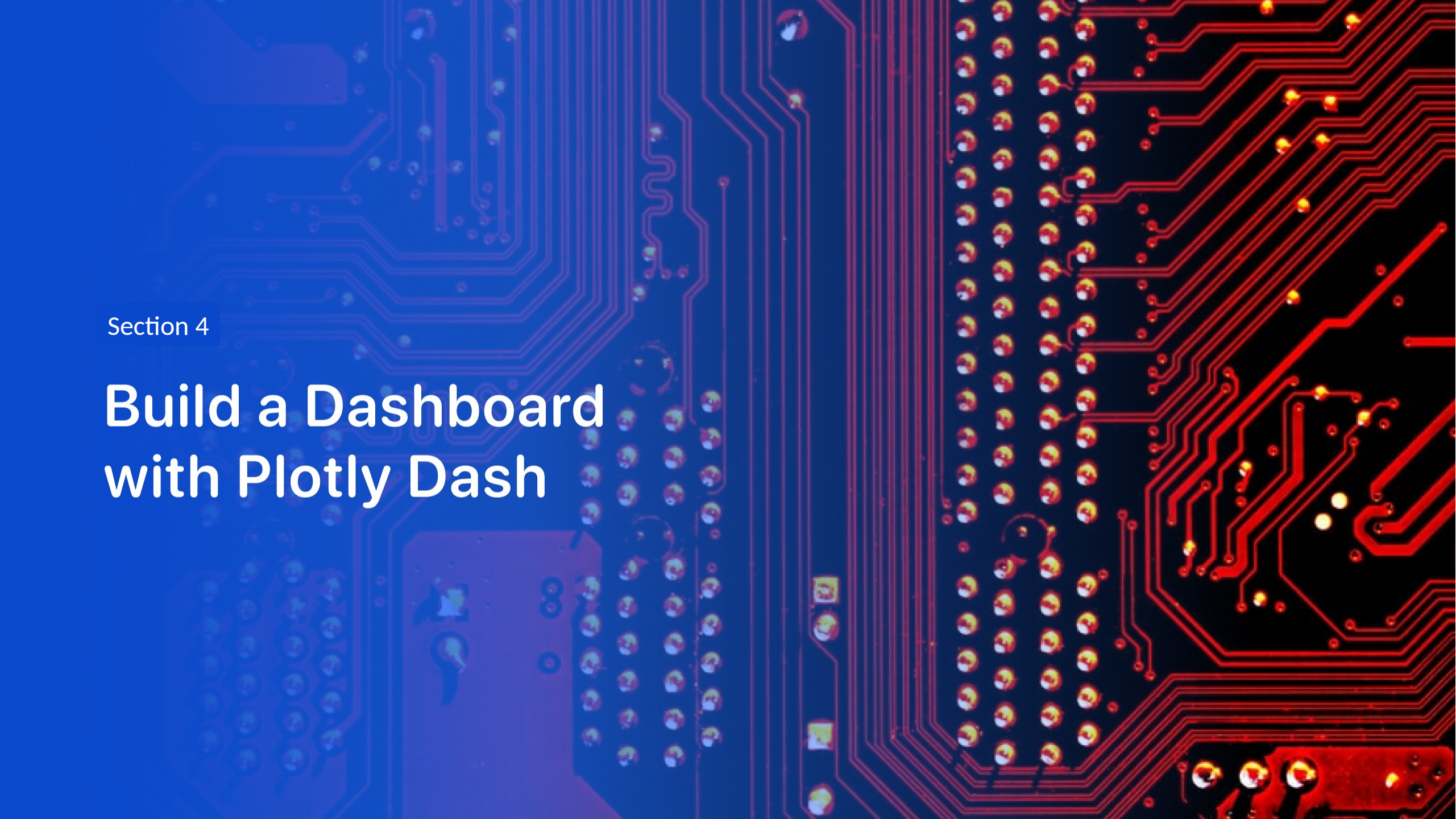
Q10



- Lines from the site to the nearest coastline, railway, highway and city show the distances calculated in the notebook
- Close to coast, rail and road: stages and heavy equipment are easy to move
- Far from cities (about 50 km or more): a failure puts few people at risk
- Distances are approximate; exact values come from the Haversine formula

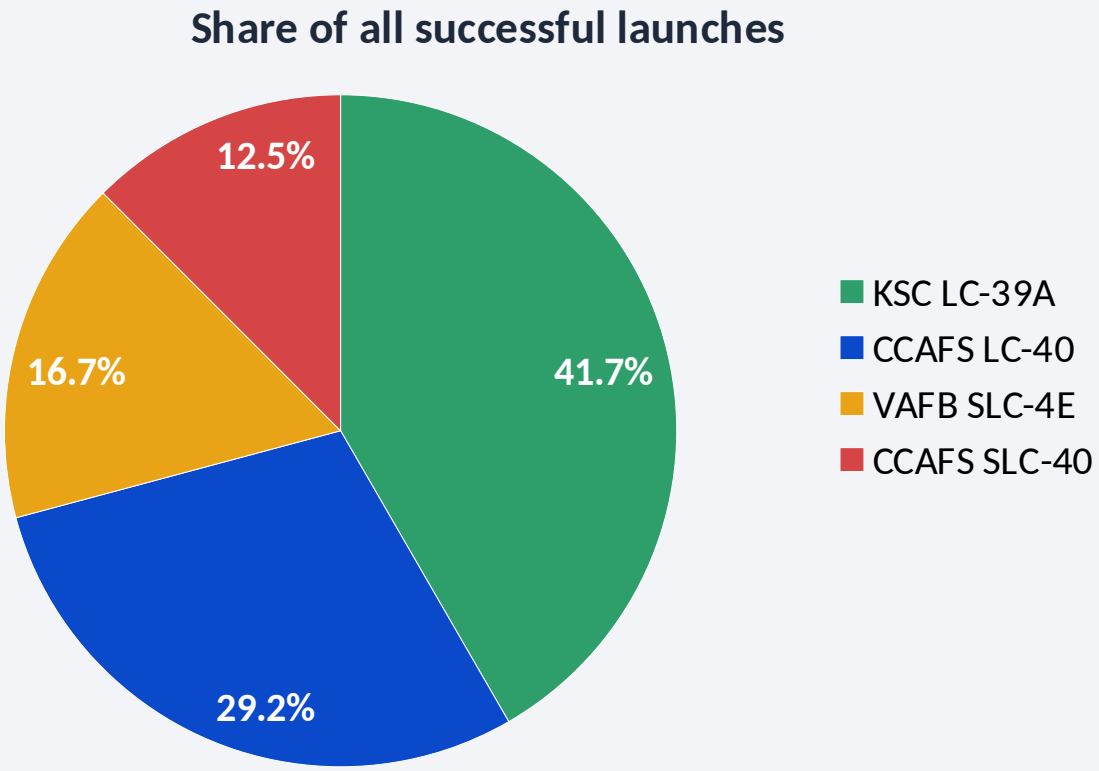
Section 4

Build a Dashboard with Plotly Dash



Launch Success Count for All Sites

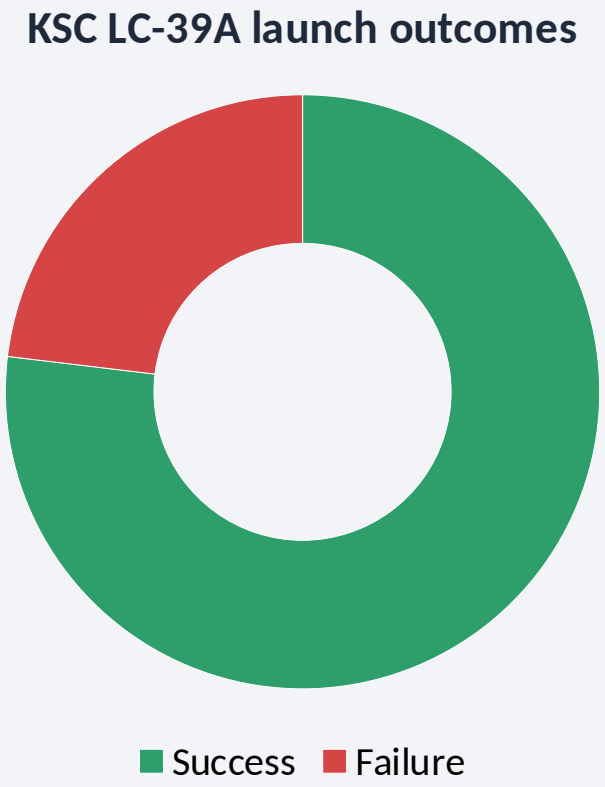
Q11



- Pie chart from the dashboard with ALL sites selected
- KSC LC-39A holds the biggest share: 41.7% of all 24 successful landings
- CCAFS LC-40 follows at 29.2%, then VAFB SLC-4E at 16.7% and CCAFS SLC-40 at 12.5%
- Takeaway: KSC LC-39A is the most productive site for successes

Highest Success Ratio: KSC LC-39A

Q11



76.9%
success (10 of 13)

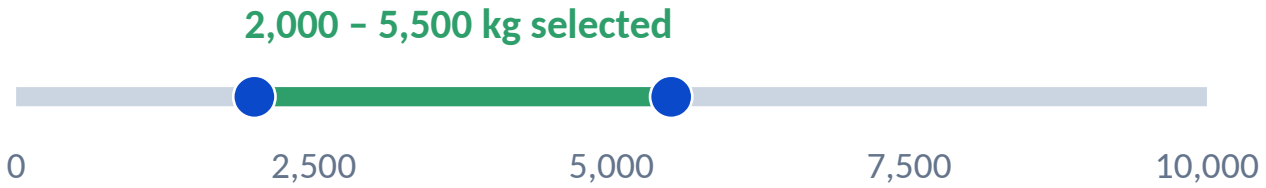
23.1%
failure (3 of 13)

- Dropdown set to KSC LC-39A shows this pie chart
- It is the site with the highest success ratio of the four
- CCAFS LC-40 (27%) and VAFB SLC-4E (40%) land far less often in this dataset

Payload vs. Outcome with the Range Slider

Q11

Payload range slider (kg)



Best payload range

About 2,000–5,500 kg has the largest share of successful landings

Best booster

FT boosters have the highest success rate in this range

What the interaction shows

- Moving the slider filters the scatter plot; points are coloured by booster version
- Low-weight payloads (0–4,000 kg) succeed more often than heavy ones
- Booster version v1.1 and B4 show the most failures
- Takeaway: payload range and booster version both help predict success

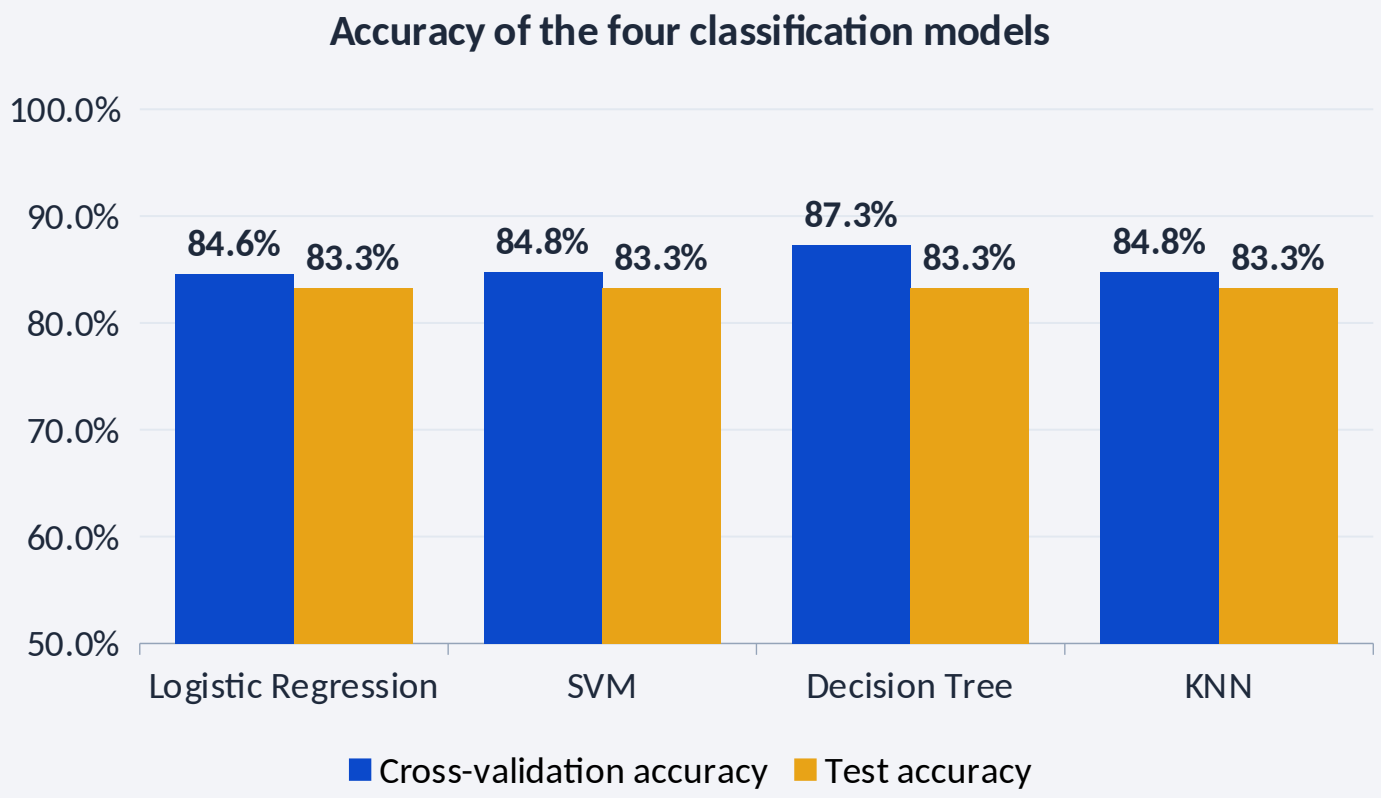
Section 5

Predictive Analysis (Classification)



Classification Accuracy

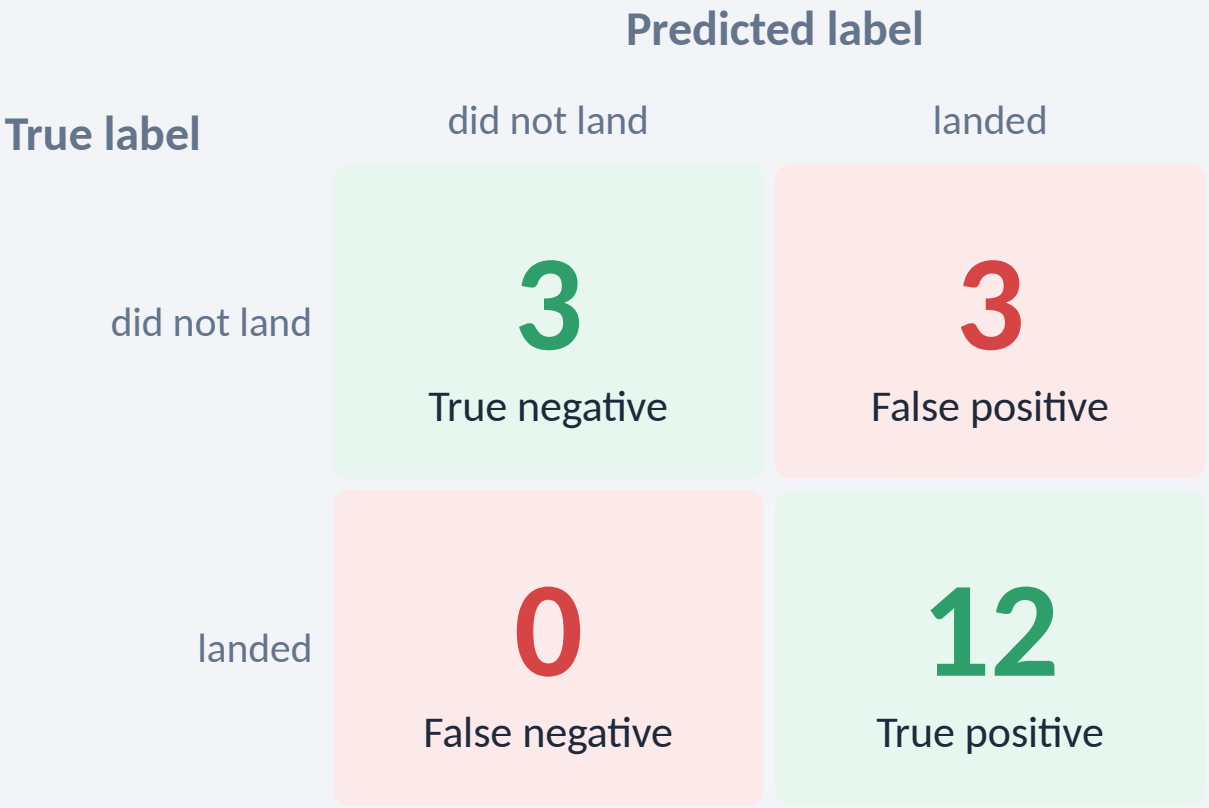
Q12



- **Best model: Decision Tree** with the highest cross-validation accuracy (87.3%)
- All four models score 83.3% on the 18 test launches
- The test set is small, so one launch moves accuracy by about 5.6 points
- Choosing by cross-validation is more reliable than the test score alone

Confusion Matrix

Q12



Decision Tree on the 18-launch test set

- 12 successful landings predicted correctly; none missed (recall 100%)
- 3 failed landings correctly identified
- 3 failures wrongly predicted as successes (false positives)
- Accuracy = 15 / 18 = 83.3%; precision for “landed” = 80%
- Main weakness: the model is optimistic and over-predicts success

Innovative Insights

Q15

0%→90%

Experience curve

Landing success went from 0% (2010–2013) to about 85–90% (2019–2020). Time and flight number act as a proxy for learning.

52% GTO

Orbit shapes the odds

High-energy GTO missions land only about 52% of the time. Low and polar orbits land 70–100%.

−60% cost

Cost lens

A $62M launch versus $165M+ is a saving of more than 60%. Each predicted landing is a direct cost estimate for a rival bidder.

3 / 18 FP

Model is optimistic

It never misses a success but gives 3 false alarms in 18. Collecting more launches would help the model learn failures.

Creativity in This Presentation

Q14



One story, not ten labs

Every notebook feeds the next step, from collection to the cost question a rival bidder would ask.

Consistent colour language

Green always means landed and red means failed, on charts, maps, the dashboard and the confusion matrix.

Custom visuals

Hand-built flowcharts, a payload range slider, a proximity diagram and a heat-mapped confusion matrix.

Plain-language takeaways

Each slide pairs the technical result with a one-line insight a non-specialist can act on.

Conclusions

Q13

- Landing success improved steadily: 0% before 2014 to roughly 85–90% in 2019–2020
- KSC LC-39A is the most successful site (76.9%); CCAFS SLC-40 launches the most (55 of 90)
- Orbit matters: SSO, HEO, GEO and ES-L1 landed every time; GTO is the hardest at about 52%
- Payloads of about 2,000–5,500 kg and Full Thrust / Block 5 boosters land most often
- A Decision Tree predicts landing with 83% test accuracy (87% cross-validated) and no missed successes
- Launch sites sit near the coast and away from cities, which supports safe operations
- Next steps: gather more launches and add engineering features to cut false positives

Answer to the business question

Yes: landing success can be predicted well enough to estimate launch cost for a competing bid.

Project repository

https://github.com/your-username/ds-capstone-spacex

All notebooks, the Dash app and CSV data are here.

Appendix

Q1

```
# Landing label (1 = landed)
df['Class'] = df['Outcome'].apply(
    lambda o: 0 if o in bad else 1)
df['Class'].mean()   # 0.667
```

```
# Tune and compare models
tree_cv = GridSearchCV(tree, p, cv=10)
tree_cv.fit(X_train, Y_train)
tree_cv.best_score_   # 0.873
tree_cv.score(X_test, Y_test)  # 0.833
```

Notebook / file	File name in repository
API data collection	jupyter-labs-spacex-data-collection-api.ipynb
Web scraping	jupyter-labs-webscraping.ipynb
Data wrangling	labs-jupyter-spacex-Data-wrangling.ipynb
EDA with SQL	jupyter-labs-eda-sql-coursera_sqllite.ipynb
EDA with visualization	edadataviz.ipynb
Folium map	lab_jupyter_launch_site_location.ipynb
Plotly Dash app	spacex_dash_app.py
Machine learning	SpaceX_Machine_Learning_Prediction_Part_5.ipynb

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Thank you!

